On the Power of Mediation in Games

Fedor Sandomirskiy Princeton University

Western, October 22, 2026

Broader agenda

Motivation

- recommendation systems and AI delegation create new channels for coordination

Solution concepts with coordination (CE, CCE, BCE, etc)

- When does coordination expand what a designer can achieve?
- Which interactions are robust to coordination by potentially malicious designers?

Related work

- *Extreme Equilibria* with Kirill Rudov and Leeat Yariv
- *Robustness to Delegation in Games* with Ben Wincelberg

Today: new results on the power of mediation, with connections to both papers

Today's question

- NE: independent, decentralized play
- CE (Aumann, 1974): a mediator sends players possibly correlated recommendations
- multiple NE already make mediation useful for coordination
- when is correlation only equilibrium selection, and when does it do more?

When is $\mathrm{conv}(\mathrm{NE}) = \mathrm{CE}$ and when are they different?

Why the gap matters

The gap $\mathrm{CE}(G) \setminus \mathrm{conv}(\mathrm{NE}(G))$ is where correlation has genuine bite:

- **Design / recommendations:** can a recommender sustain outcomes beyond decentralized play?
- **Communication:** is pre-play talk just coordination on which NE to play?
  (Bárány, 1992; Forges, 1990; Lehrer and Sorin, 1997; Ben-Porath, 1998; Gerardi, 2004)
- **Learning:** do no-regret or calibrated dynamics recover decentralized predictions?
  (Foster and Vohra, 1997; Fudenberg and Levine, 1999; Hart and Mas-Colell, 2000)
- **Computation:** when does a tractable LP describe a decentralized prediction?
  (Papadimitriou and Roughgarden, 2008)
- **Decision theory:** when does Bayesian rationality collapse to Nash logic?
  (Aumann, 1987)

What is known

CE can be strictly larger than $\mathrm{conv}(\mathrm{NE})$:

- the original 2×2 example (Aumann, 1974); all non-degenerate 2×2 games with two pure NE (Calvó-Armengol, 2006); mediation value in 2×2 games (Ashlagi, Monderer, and Tennenholtz, 2008); all-pay auctions (Pavlov, 2023)

What is known

CE can be strictly larger than conv(NE):

- the original 2×2 example (Aumann, 1974); all non-degenerate 2×2 games with two pure NE (Calvó-Armengol, 2006); mediation value in 2×2 games (Ashlagi, Monderer, and Tennenholtz, 2008); all-pay auctions (Pavlov, 2023)

CE can collapse to conv(NE):

- concave potential games (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025); zero-sum games (Forges, 1990; Viossat, 2006; Kattwinkel, Niemeyer, Preusser, and Winter, 2022; Kartik, Squintani, and Tinn, 2024); first-price auctions (Feldman, Lucier, and Nisan, 2016); non-atomic congestion games (Koessler, Scarsini, and Tomala, 2025)

What is known

CE can be strictly larger than $\mathrm{conv}(\mathrm{NE})$:

- the original 2×2 example (Aumann, 1974); all non-degenerate 2×2 games with two pure NE (Calvó-Armengol, 2006); mediation value in 2×2 games (Ashlagi, Monderer, and Tennenholtz, 2008); all-pay auctions (Pavlov, 2023)

CE can collapse to $\mathrm{conv}(\mathrm{NE})$:

- concave potential games (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025); zero-sum games (Forges, 1990; Viossat, 2006; Kattwinkel, Niemeyer, Preusser, and Winter, 2022; Kartik, Squintani, and Tinn, 2024); first-price auctions (Feldman, Lucier, and Nisan, 2016); non-atomic congestion games (Koessler, Scarsini, and Tomala, 2025)

Maximal collapse, $\mathrm{CE} = \{$point$\}$:

- an open set (Viossat, 2008), robust to incomplete information (Kajii and Morris, 1997; Einy, Haimanko, and Lagziel, 2022), characterization of unique CCE (Sandomirskiy and Wincelberg, 2026)

What is known

CE can be strictly larger than $\mathrm{conv}(\mathrm{NE})$:

- the original 2×2 example (Aumann, 1974); all non-degenerate 2×2 games with two pure NE (Calvó-Armengol, 2006); mediation value in 2×2 games (Ashlagi, Monderer, and Tennenholtz, 2008); all-pay auctions (Pavlov, 2023)

CE can collapse to $\mathrm{conv}(\mathrm{NE})$:

- concave potential games (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025); zero-sum games (Forges, 1990; Viossat, 2006; Kattwinkel, Niemeyer, Preusser, and Winter, 2022; Kartik, Squintani, and Tinn, 2024); first-price auctions (Feldman, Lucier, and Nisan, 2016); non-atomic congestion games (Koessler, Scarsini, and Tomala, 2025)

Maximal collapse, $\mathrm{CE} = \{$point$\}$:

- an open set (Viossat, 2008), robust to incomplete information (Kajii and Morris, 1997; Einy, Haimanko, and Lagziel, 2022), characterization of unique CCE (Sandomirskiy and Wincelberg, 2026)

Extremality of a NE inside CE:

- always extreme in two-player games (Cripps, 1995; Evangelista and Raghavan, 1996; Canovas, Hansen, and Jaumard, 1999); non-extreme iff 3+ players mix (Rudov, Sandomirskiy, and Yariv, 2025)

Outline

1. Model: games, CE, NE, regularity
2. Characterization of games with $\mathrm{CE}(G) \setminus \mathrm{conv}(\mathrm{NE}(G)) \neq \emptyset$
3. How to construct points outside the hull
   - Tools: extremality, complex-valued Nash equilibria, and index theory
4. Games with a unique CE and CCE

Model

Games and equilibrium concepts

- finite normal-form games $G = (A, u)$, where $A = A_1 \times \cdots \times A_n$ and $u \colon A \to \mathbb{R}^n$

Games and equilibrium concepts

- finite normal-form games $G = (A, u)$, where $A = A_1 \times \cdots \times A_n$ and $u \colon A \to \mathbb{R}^n$
- a mediator draws recommendations $a = (a_1, \ldots, a_n)$ from $\mu \in \Delta(A)$

Games and equilibrium concepts

- finite normal-form games $G = (A, u)$, where $A = A_1 \times \cdots \times A_n$ and $u \colon A \to \mathbb{R}^n$
- a mediator draws recommendations $a = (a_1, \ldots, a_n)$ from $\mu \in \Delta(A)$

Correlated equilibrium (Aumann, 1974)

$\mu \in \Delta(A)$ is a CE if every recommendation is self-enforcing: for all i and all $a_i, a_i' \in A_i$

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) \big[u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) \big] \;\geq\; 0$$

Games and equilibrium concepts

- finite normal-form games $G = (A, u)$, where $A = A_1 \times \cdots \times A_n$ and $u \colon A \to \mathbb{R}^n$
- a mediator draws recommendations $a = (a_1, \ldots, a_n)$ from $\mu \in \Delta(A)$

Correlated equilibrium (Aumann, 1974)

$\mu \in \Delta(A)$ is a CE if every recommendation is self-enforcing: for all i and all $a_i, a_i' \in A_i$

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) \big[u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) \big] \;\geq\; 0$$

- $\mathrm{CE}(G)$ is a polytope: finitely many linear inequalities on $\Delta(A)$
- an NE is a CE that is a product: $\mu = \mu_1 \times \cdots \times \mu_n$
- $\mathrm{conv}(\mathrm{NE})$: outcomes from coordinating equilibrium selection

Regularity

- pathological games blur the question
- if $u_i \equiv 0$ for all i, then $\mathrm{CE} = \mathrm{conv}(\mathrm{NE}) = \Delta(A)$
- correlation adds nothing for the wrong reason: there are too many NE

Regularity

- pathological games blur the question
- if $u_i \equiv 0$ for all i, then $\mathrm{CE} = \mathrm{conv}(\mathrm{NE}) = \Delta(A)$
- correlation adds nothing for the wrong reason: there are too many NE

Regular game

G is regular if it has finitely many NE and each is

- **quasi-strict**: deviations outside the support are strictly unprofitable
- **nondegenerate**: the support-indifference Jacobian is invertible

Regularity

- pathological games blur the question
- if $u_i \equiv 0$ for all i, then $\mathrm{CE} = \mathrm{conv}(\mathrm{NE}) = \Delta(A)$
- correlation adds nothing for the wrong reason: there are too many NE

Regular game

G is regular if it has finitely many NE and each is

- **quasi-strict**: deviations outside the support are strictly unprofitable
- **nondegenerate**: the support-indifference Jacobian is invertible

- each NE is robust: varies smoothly under small payoff perturbations
- Harsanyi (1973): all games except a measure-zero set are regular
- often quasi-strictness will be enough

Characterization

Main result: a sharp dichotomy

Theorem

Let G be regular. Then exactly one of the following holds:

$$\underbrace{\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset}_{\text{mediation has real bite}} \quad \text{OR} \quad \underbrace{\mathrm{CE} = \{\text{point}\}}_{\text{mediation powerless}}$$

Main result: a sharp dichotomy

Theorem

Let G be regular. Then exactly one of the following holds:

$$\underbrace{\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset}_{\text{mediation has real bite}} \quad \text{OR} \quad \underbrace{\mathrm{CE} = \{\text{point}\}}_{\text{mediation powerless}}$$

- An easy-to-apply corollary: $\#\mathrm{NE} \geq 2 \implies \mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset$

Main result: a sharp dichotomy

Theorem

Let G be regular. Then exactly one of the following holds:

$$\underbrace{\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset}_{\text{mediation has real bite}} \quad \text{OR} \quad \underbrace{\mathrm{CE} = \{\text{point}\}}_{\text{mediation powerless}}$$

- An easy-to-apply corollary: $\#\mathrm{NE} \geq 2 \implies \mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset$

Interpretation

- communication and recommendations expand outcomes whenever coordination failure is possible
- learning or LP solving are are guaranteed to reach a decentralized outcome only when CE is a singleton

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

- complete-information reduction: player i's action is a decision rule $s_i \colon T_i \to A_i$

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

- complete-information reduction: player i's action is a decision rule $s_i \colon T_i \to A_i$
- NE of the reduced game $\simeq$ Bayes-NE

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

- complete-information reduction: player i's action is a decision rule $s_i \colon T_i \to A_i$
- NE of the reduced game $\simeq$ Bayes-NE
- CE of the reduced game $\simeq$ ex-ante state-independent CE

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

- complete-information reduction: player i's action is a decision rule $s_i \colon T_i \to A_i$
- NE of the reduced game $\simeq$ Bayes-NE
- CE of the reduced game $\simeq$ ex-ante state-independent CE
- the reduced game is finite $\Rightarrow$ the theorem applies

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

- complete-information reduction: player i's action is a decision rule $s_i \colon T_i \to A_i$
- NE of the reduced game $\simeq$ Bayes-NE
- CE of the reduced game $\simeq$ ex-ante state-independent CE
- the reduced game is finite $\Rightarrow$ the theorem applies

Corollary

For a regular Bayesian game, either $\mathrm{CE} \setminus \mathrm{conv}(\text{Bayes-}\mathrm{NE}) \neq \emptyset$ OR $\mathrm{CE} = \{\text{point}\}$

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

- complete-information reduction: player i's action is a decision rule $s_i \colon T_i \to A_i$
- NE of the reduced game $\simeq$ Bayes-NE
- CE of the reduced game $\simeq$ ex-ante state-independent CE
- the reduced game is finite $\Rightarrow$ the theorem applies

Corollary

For a regular Bayesian game, either $\mathrm{CE} \setminus \mathrm{conv}(\text{Bayes-NE}) \neq \emptyset$ OR $\mathrm{CE} = \{\text{point}\}$

- CE (mediator knows nothing) $\subseteq$ communication equilibria (players report types) $\subseteq$ BCE (mediator omnipotent)

Extension to Bayesian games

States Θ, finite types T_i, prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

- complete-information reduction: player i's action is a decision rule $s_i \colon T_i \to A_i$
- NE of the reduced game $\simeq$ Bayes-NE
- CE of the reduced game $\simeq$ ex-ante state-independent CE
- the reduced game is finite $\Rightarrow$ the theorem applies

Corollary

For a regular Bayesian game, either $\mathrm{CE} \setminus \mathrm{conv}(\text{Bayes-NE}) \neq \emptyset$ OR $\mathrm{CE} = \{\text{point}\}$

- CE (mediator knows nothing) $\subseteq$ communication equilibria (players report types) $\subseteq$ BCE (mediator omnipotent)
- $\#\text{Bayes-NE} \geq 2 \implies$ ex-ante state-independent CE, communication equilibria, and BCE all strictly contain $\mathrm{conv}(\text{Bayes-NE})$

How to construct points outside the hull

Where do points outside the hull come from?

Goal: regular G and CE not a single point $\Rightarrow$ $\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset$

Where do points outside the hull come from?

Goal: regular G and CE not a single point $\Rightarrow$ $\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset$

- **Trivial case:** a single NE. Then $\mathrm{conv}(\mathrm{NE})$ is a point, so if CE is not a singleton the inclusion is automatically strict

Where do points outside the hull come from?

Goal: regular G and CE not a single point $\Rightarrow$ $\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset$

- **Trivial case:** a single NE. Then $\mathrm{conv}(\mathrm{NE})$ is a point, so if CE is not a singleton the inclusion is automatically strict
- **So assume** ≥ 2 **Nash equilibria** We must construct a CE outside the Nash hull

Where do points outside the hull come from?

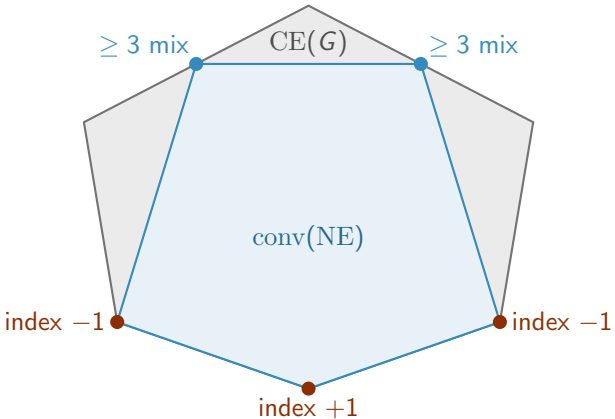
Goal: regular G and CE not a single point $\Rightarrow$ $\mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE}) \neq \emptyset$

- **Trivial case:** a single NE. Then $\mathrm{conv}(\mathrm{NE})$ is a point, so if CE is not a singleton the inclusion is automatically strict
- **So assume** ≥ 2 **Nash equilibria** We must construct a CE outside the Nash hull

Two regimes require different arguments:

some NE **has** ≥ 3 **mixers:**	extremality and and complex Nash equilibria
every NE **has** ≤ 2 **mixers:**	index and anti-Nash equilibria

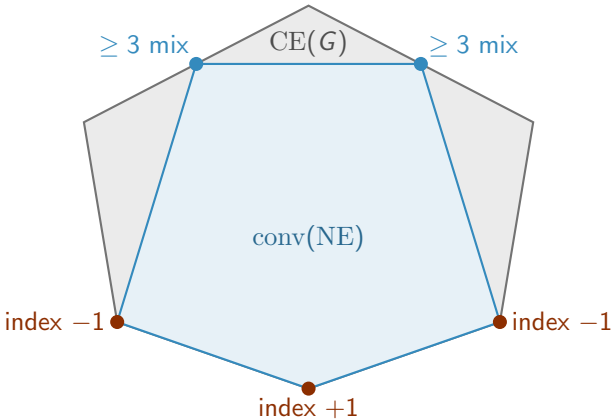
The two ways to leave the hull



the three •: NE with ≤ 2 mixers

- NE with ≥ 3 mixers sit inside faces: corners of CE escape the hull

The two ways to leave the hull



the three •: NE with ≤ 2 mixers

- NE with ≥ 3 mixers sit inside faces: corners of CE escape the hull
- NE with ≤ 2 mixers sit at vertices of CE (extreme)
  - index -1 $\Rightarrow$ admit a perturbation exiting the hull

Equilibria with three or more mixing players

Equilibria with three or more mixing players

- Rudov, Sandomirskiy, and Yariv (2025): A quasi-strict NE with ≥ 3 mixing players is not extreme
- **Idea:** more perturbation directions than incentive constraints

Details

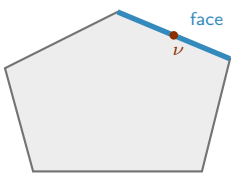
Equilibria with three or more mixing players

- Rudov, Sandomirskiy, and Yariv (2025): A quasi-strict NE with ≥ 3 mixing players is not extreme
- **Idea:** more perturbation directions than incentive constraints

Details

Proposition

Let ν be a quasi-strict NE with ≥ 3 mixing players. Then no extreme point of the CE-face carrying ν is a quasi-strict NE

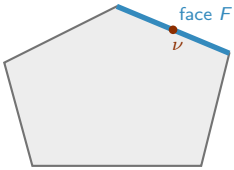


- $\Rightarrow$ those extreme points lie outside $\mathrm{conv}(\mathrm{NE})$
- in fact, all boundary points of this face are outside $\mathrm{conv}(\mathrm{NE})$

Proof sketch

Proposition

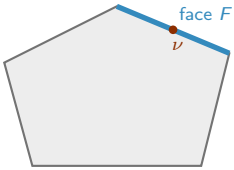
Let ν be a quasi-strict NE *with ≥ 3 mixing players. Then no extreme point of the* CE*-face carrying ν is a quasi-strict* NE



Proof sketch

Proposition

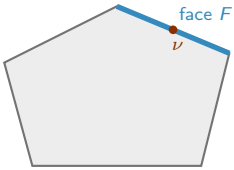
Let ν be a quasi-strict NE *with ≥ 3 mixing players. Then no extreme point of the* CE*-face carrying ν is a quasi-strict* NE



Proof sketch:

- let F be the minimal CE-face containing ν

Proof sketch



Proposition

Let ν be a quasi-strict NE *with ≥ 3 mixing players. Then no extreme point of the* CE*-face carrying ν is a quasi-strict* NE

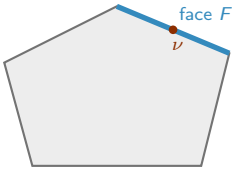
Proof sketch:

- let F be the minimal CE-face containing ν
- suppose an extreme point $e \in F$ is a quasi-strict NE

Proof sketch

Proposition

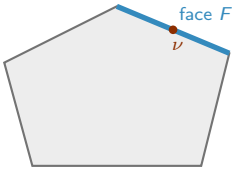
Let ν be a quasi-strict NE *with ≥ 3 mixing players. Then no extreme point of the* CE*-face carrying ν is a quasi-strict* NE



Proof sketch:

- let F be the minimal CE-face containing ν
- suppose an extreme point $e \in F$ is a quasi-strict NE
- the constraints binding at ν also bind throughout F

Proof sketch



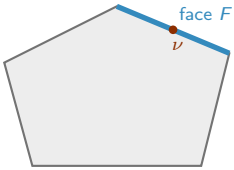
Proposition

Let ν be a quasi-strict NE with ≥ 3 mixing players. Then no extreme point of the CE-face carrying ν is a quasi-strict NE

Proof sketch:

- let F be the minimal CE-face containing ν
- suppose an extreme point $e \in F$ is a quasi-strict NE
- the constraints binding at ν also bind throughout F
- $\Rightarrow \nu(a) = 0 \Rightarrow e(a) = 0$, that is $\mathrm{supp}(e) \subseteq \mathrm{supp}(\nu)$

Proof sketch



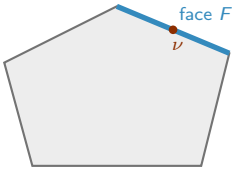
Proposition

Let ν be a quasi-strict NE with ≥ 3 mixing players. Then no extreme point of the CE-face carrying ν is a quasi-strict NE

Proof sketch:

- let F be the minimal CE-face containing ν
- suppose an extreme point $e \in F$ is a quasi-strict NE
- the constraints binding at ν also bind throughout F
- $\Rightarrow \nu(a) = 0 \Rightarrow e(a) = 0$, that is $\mathrm{supp}(e) \subseteq \mathrm{supp}(\nu)$
- at ν players are indifferent over $\mathrm{supp}(\nu) \Rightarrow$ at e all players are indifferent over $\mathrm{supp}(\nu)$

Proof sketch



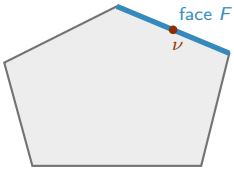
Proposition

Let ν be a quasi-strict NE with ≥ 3 mixing players. Then no extreme point of the CE-face carrying ν is a quasi-strict NE

Proof sketch:

- let F be the minimal CE-face containing ν
- suppose an extreme point $e \in F$ is a quasi-strict NE
- the constraints binding at ν also bind throughout F
- $\Rightarrow \nu(a) = 0 \Rightarrow e(a) = 0$, that is $\mathrm{supp}(e) \subseteq \mathrm{supp}(\nu)$
- at ν players are indifferent over $\mathrm{supp}(\nu) \Rightarrow$ at e all players are indifferent over $\mathrm{supp}(\nu)$
- quasi-strictness $\Rightarrow$ all these actions are used $\Rightarrow \mathrm{supp}(e) = \mathrm{supp}(\nu)$

Proof sketch



Proposition

Let ν be a quasi-strict NE with ≥ 3 mixing players. Then no extreme point of the CE-face carrying ν is a quasi-strict NE

Proof sketch:

- let F be the minimal CE-face containing ν
- suppose an extreme point $e \in F$ is a quasi-strict NE
- the constraints binding at ν also bind throughout F
- $\Rightarrow \nu(a) = 0 \Rightarrow e(a) = 0$, that is $\mathrm{supp}(e) \subseteq \mathrm{supp}(\nu)$
- at ν players are indifferent over $\mathrm{supp}(\nu) \Rightarrow$ at e all players are indifferent over $\mathrm{supp}(\nu)$
- quasi-strictness $\Rightarrow$ all these actions are used $\Rightarrow \mathrm{supp}(e) = \mathrm{supp}(\nu)$
- so e also has ≥ 3 mixers, contradicting extremality

□

Complex Nash construction

Complex-valued Nash equilibria

Fix a NE $\nu = \nu_1 \times \cdots \times \nu_n$ with ≥ 3 mixers in a regular game

On $S = \mathrm{supp}(\nu)$, indifference requires

$$u_i(a_i, \nu_{-i}) - u_i(b_i, \nu_{-i}) = 0, \qquad a_i, b_i \in S_i$$

Complex-valued Nash equilibria

Fix a NE $\nu = \nu_1 \times \cdots \times \nu_n$ with ≥ 3 mixers in a regular game

On $S = \mathrm{supp}(\nu)$, indifference requires

$$u_i(a_i, \nu_{-i}) - u_i(b_i, \nu_{-i}) = 0, \qquad a_i, b_i \in S_i$$

Complex Nash equilibrium

Find a product solution $\nu' = \nu'_1 \times \cdots \times \nu'_n$ on S, not proportional to ν, with $\nu'_i(a_i) \in \mathbb{C}$

Complex-valued Nash equilibria

Fix a NE $\nu = \nu_1 \times \cdots \times \nu_n$ with ≥ 3 mixers in a regular game

On $S = \mathrm{supp}(\nu)$, indifference requires

$$u_i(a_i, \nu_{-i}) - u_i(b_i, \nu_{-i}) = 0, \qquad a_i, b_i \in S_i$$

Complex Nash equilibrium

Find a product solution $\nu' = \nu'_1 \times \cdots \times \nu'_n$ on S, not proportional to ν, with $\nu'_i(a_i) \in \mathbb{C}$

Idea

- complex solutions satisfy the same indifference equations
- $\Rightarrow$ taking real parts gives a signed measure satisfying indifference
- $\Rightarrow$ a perturbation direction at the original equilibrium

Real perturbations from complex equilibria

Construction as an alternative to solving an LP

Set $h = \nu' - \nu'(A)\nu$ and rescale ν' so $\operatorname{Re} h \neq 0$. Define

$$\mu_\varepsilon = \nu + \varepsilon \operatorname{Re} h$$

Then μ_ε has mass 1 and, by quasi-strictness, is a CE for small $|\varepsilon|$

Real perturbations from complex equilibria

Construction as an alternative to solving an LP

Set $h = \nu' - \nu'(A)\nu$ and rescale ν' so $\operatorname{Re} h \neq 0$. Define

$$\mu_\varepsilon = \nu + \varepsilon \operatorname{Re} h$$

Then μ_ε has mass 1 and, by quasi-strictness, is a CE for small $|\varepsilon|$

Proposition

In a regular game and ν with ≥ 3 mixers, this construction reaches a CE outside the Nash hull

Real perturbations from complex equilibria

Construction as an alternative to solving an LP

Set $h = \nu' - \nu'(A)\nu$ and rescale ν' so $\operatorname{Re} h \neq 0$. Define

$$\mu_\varepsilon = \nu + \varepsilon \operatorname{Re} h$$

Then μ_ε has mass 1 and, by quasi-strictness, is a CE for small $|\varepsilon|$

Proposition

In a regular game and ν with ≥ 3 mixers, this construction reaches a CE outside the Nash hull

Idea of the proof

- the boundary of the minimal CE-face carrying ν is outside the hull
- generic root counts guarantee another complex solution (McKelvey and McLennan, 1997): for a given support S,

$$\#\text{complex roots} \geq (\#\text{mixing players} - 1)! \geq 2$$

  - Bernstein–Kushnirenko–Khovanskii theorem (Kaveh and Khovanskii, 2018)

Details

Example: four participants

A participation-coordination game with four players

- participation costs 1 and pays 8 if everyone participates
- abstaining gives payoff 0
- three NE: all abstain, all participate, and independent participation with probability $1/2$

Example: four participants

A participation-coordination game with four players

- participation costs 1 and pays 8 if everyone participates
- abstaining gives payoff 0
- three NE: all abstain, all participate, and independent participation with probability $1/2$

The fully mixed indifference conditions give a common participation probability $p = \nu_i(1)$

$$8p^3 = 1, \qquad p = \frac{1}{2},\ \frac{\omega}{2},\ \frac{\overline{\omega}}{2}, \qquad \omega = e^{2\pi \mathrm{i}/3}$$

Real solution: $\nu = (1/2, 1/2) \times \cdots \times (1/2, 1/2)$

Complex solution: $\nu' = \nu'_1 \times \cdots \times \nu'_4$, with $\nu'_i = (1 - \omega/2, \omega/2)$

The resulting mediator

Here $\nu'(A) = 1$, so $h = \nu' - \nu$ and the negative endpoint is

$$\mu = \nu - \frac{2}{21}\operatorname{Re} h, \qquad \varepsilon = -\frac{2}{21}$$

number of participants	0	1	2	3	4
probability	0	1/14	9/14	3/14	1/14

Choose a subset of this size uniformly and recommend participation to its members

The resulting mediator

Here $\nu'(A) = 1$, so $h = \nu' - \nu$ and the negative endpoint is

$$\mu = \nu - \frac{2}{21}\operatorname{Re} h, \qquad \varepsilon = -\frac{2}{21}$$

number of participants	0	1	2	3	4
probability	0	1/14	9/14	3/14	1/14

Choose a subset of this size uniformly and recommend participation to its members

- after either recommendation, all others participate with probability $1/8$: obedience is optimal
- nonunanimous play has probability $13/14 > \underbrace{7/8}_{\text{maximum over Nash mixtures}}$

Two constructions at the same equilibrium

In a regular game, at any NE with at ≥ 3 mixers:

Linear dimension-counting approach

Move to the boundary of the minimal CE face, for example by solving an LP

Complex Nash approach

Use another complex solution to construct a direction, then move to the boundary

Surprisingly, both routes work simultaneously!

At most two mixers at every equilibrium

At most two mixers at every equilibrium

- suppose every NE has at most two mixers

At most two mixers at every equilibrium

- suppose every NE has at most two mixers
- under regularity, each NE is pure or has two mixers with equal support sizes

At most two mixers at every equilibrium

- suppose every NE has at most two mixers
- under regularity, each NE is pure or has two mixers with equal support sizes
- all such equilibria are extreme in CE (Cripps, 1995; Evangelista and Raghavan, 1996; Canovas, Hansen, and Jaumard, 1999; Rudov, Sandomirskiy, and Yariv, 2025)

At most two mixers at every equilibrium

- suppose every NE has at most two mixers
- under regularity, each NE is pure or has two mixers with equal support sizes
- all such equilibria are extreme in CE (Cripps, 1995; Evangelista and Raghavan, 1996; Canovas, Hansen, and Jaumard, 1999; Rudov, Sandomirskiy, and Yariv, 2025)
- $\Rightarrow$ the previous face construction cannot work

At most two mixers at every equilibrium

- suppose every NE has at most two mixers
- under regularity, each NE is pure or has two mixers with equal support sizes
- all such equilibria are extreme in CE (Cripps, 1995; Evangelista and Raghavan, 1996; Canovas, Hansen, and Jaumard, 1999; Rudov, Sandomirskiy, and Yariv, 2025)
- $\Rightarrow$ the previous face construction cannot work

Two ideas to the rescue

1. anti-Nash: equilibria after payoff reversal $u \mapsto -u$ can be used to construct CE (Cripps, 1995)
2. index theory: the construction works at equilibria of index -1

Crash course in index theory

For a NE ν, $\mathrm{index}_u(\nu) = \mathrm{sgn}\det J(\nu)$, where J is the Jacobian of the indifference equations

Crash course in index theory

For a NE ν, $\quad \mathrm{index}_u(\nu) = \operatorname{sgn}\det J(\nu)$, $\quad$ where J is the Jacobian of the indifference equations

- player i randomizes over m_i actions $\Rightarrow$ the $m_i - 1$ indifference equations:

$$u_i(b_i, \nu_{-i}) - u_i(a_i, \nu_{-i}) = 0, \qquad a_i \in \operatorname{supp}(\nu_i) \setminus \{b_i\} \quad \text{for some fixed } b_i$$

Crash course in index theory

For a NE ν, $\quad \mathrm{index}_u(\nu) = \operatorname{sgn}\det J(\nu)$, $\quad$ where J is the Jacobian of the indifference equations

- player i randomizes over m_i actions $\Rightarrow$ the $m_i - 1$ indifference equations:
$$u_i(b_i, \nu_{-i}) - u_i(a_i, \nu_{-i}) = 0, \qquad a_i \in \operatorname{supp}(\nu_i) \setminus \{b_i\} \quad \text{for some fixed } b_i$$

- $J(\nu)$ is the Jacobian of all these, a $d \times d$ matrix, $d = \sum_i (m_i - 1)$, with zero diagonal blocks
$$J = \begin{pmatrix} 0 & * & * \\ * & 0 & * \\ * & * & 0 \end{pmatrix}$$
(own indifference does not depend on own mixing)

Crash course in index theory

For a NE ν, $\quad \mathrm{index}_u(\nu) = \operatorname{sgn}\det J(\nu)$, $\quad$ where J is the Jacobian of the indifference equations

- player i randomizes over m_i actions $\Rightarrow$ the $m_i - 1$ indifference equations:

$$u_i(b_i, \nu_{-i}) - u_i(a_i, \nu_{-i}) = 0, \qquad a_i \in \operatorname{supp}(\nu_i) \setminus \{b_i\} \quad \text{for some fixed } b_i$$

- $J(\nu)$ is the Jacobian of all these, a $d \times d$ matrix, $d = \sum_i (m_i - 1)$, with zero diagonal blocks

$$J = \begin{pmatrix} 0 & * & * \\ * & 0 & * \\ * & * & 0 \end{pmatrix} \qquad \text{(own indifference does not depend on own mixing)}$$

Properties

- regular $\Rightarrow \det J \neq 0$, so $\mathrm{index} = \pm 1$; pure $\Rightarrow \mathrm{index} = +1$ (0×0 determinant)

Crash course in index theory

For a NE ν, $\quad \mathrm{index}_u(\nu) = \operatorname{sgn}\det J(\nu)$, $\quad$ where J is the Jacobian of the indifference equations

- player i randomizes over m_i actions $\Rightarrow$ the $m_i - 1$ indifference equations:

$$u_i(b_i, \nu_{-i}) - u_i(a_i, \nu_{-i}) = 0, \qquad a_i \in \operatorname{supp}(\nu_i) \setminus \{b_i\} \quad \text{for some fixed } b_i$$

- $J(\nu)$ is the Jacobian of all these, a $d \times d$ matrix, $d = \sum_i (m_i - 1)$, with zero diagonal blocks

$$J = \begin{pmatrix} 0 & * & * \\ * & 0 & * \\ * & * & 0 \end{pmatrix}$$

(own indifference does not depend on own mixing)

Properties

- regular $\Rightarrow \det J \neq 0$, so $\mathrm{index} = \pm 1$; pure $\Rightarrow \mathrm{index} = +1$ (0×0 determinant)
- restricting to the support preserves index

Crash course in index theory

For a NE ν, $\quad \mathrm{index}_u(\nu) = \operatorname{sgn}\det J(\nu)$, $\quad$ where J is the Jacobian of the indifference equations

- player i randomizes over m_i actions $\Rightarrow$ the $m_i - 1$ indifference equations:

$$u_i(b_i, \nu_{-i}) - u_i(a_i, \nu_{-i}) = 0, \qquad a_i \in \operatorname{supp}(\nu_i) \setminus \{b_i\} \quad \text{for some fixed } b_i$$

- $J(\nu)$ is the Jacobian of all these, a $d \times d$ matrix, $d = \sum_i (m_i - 1)$, with zero diagonal blocks

$$J = \begin{pmatrix} 0 & * & * \\ * & 0 & * \\ * & * & 0 \end{pmatrix}$$

(own indifference does not depend on own mixing)

Properties

- regular $\Rightarrow \det J \neq 0$, so $\mathrm{index} = \pm 1$; pure $\Rightarrow \mathrm{index} = +1$ (0×0 determinant)
- restricting to the support preserves index
- after restriction, payoff reversal gives $\mathrm{index}_{-u}(\nu) = (-1)^d \,\mathrm{index}_u(\nu)$

Crash course in index theory

For a NE ν, $\quad \mathrm{index}_u(\nu) = \mathrm{sgn}\det J(\nu)$, $\quad$ where J is the Jacobian of the indifference equations

- player i randomizes over m_i actions $\Rightarrow$ the $m_i - 1$ indifference equations:

$$u_i(b_i, \nu_{-i}) - u_i(a_i, \nu_{-i}) = 0, \qquad a_i \in \mathrm{supp}(\nu_i) \setminus \{b_i\} \quad \text{for some fixed } b_i$$

- $J(\nu)$ is the Jacobian of all these, a $d \times d$ matrix, $d = \sum_i (m_i - 1)$, with zero diagonal blocks

$$J = \begin{pmatrix} 0 & * & * \\ * & 0 & * \\ * & * & 0 \end{pmatrix} \qquad \text{(own indifference does not depend on own mixing)}$$

Properties

- regular $\Rightarrow \det J \neq 0$, so $\mathrm{index} = \pm 1$; pure $\Rightarrow \mathrm{index} = +1$ (0×0 determinant)
- restricting to the support preserves index
- after restriction, payoff reversal gives $\mathrm{index}_{-u}(\nu) = (-1)^d \,\mathrm{index}_u(\nu)$
- two mixers: $d = 2(m-1)$ is even $\Rightarrow \mathrm{index}_{-u}(\nu) = \mathrm{index}_u(\nu)$
- $\mathrm{index}_{-u}(\nu) \Rightarrow$ dynamical instability

Crash course in index theory

Index theorem (Gül, Pearce, and Stacchetti, 1993; Ritzberger, 1994)

For a regular game, $\sum_{\nu \in \mathrm{NE}} \mathrm{index}(\nu) = +1$

one can also assign an index to equilibrium components when regularity fails, but we won't need it

Crash course in index theory

Index theorem (Gül, Pearce, and Stacchetti, 1993; Ritzberger, 1994)

For a regular game, $\sum_{\nu \in \mathrm{NE}} \mathrm{index}(\nu) = +1$

one can also assign an index to equilibrium components when regularity fails, but we won't need it

Corollaries

- $\#\{\text{mixed NE}\} \geq \#\{\text{pure NE}\} - 1$ (Gül, Pearce, and Stacchetti, 1993)

Crash course in index theory

Index theorem (Gül, Pearce, and Stacchetti, 1993; Ritzberger, 1994)

For a regular game, $\sum_{\nu \in \mathrm{NE}} \mathrm{index}(\nu) = +1$

one can also assign an index to equilibrium components when regularity fails, but we won't need it

Corollaries

- $\#\{\text{mixed NE}\} \geq \#\{\text{pure NE}\} - 1$ (Gül, Pearce, and Stacchetti, 1993)
- ≥ 2 NE $\Rightarrow$ some equilibrium has index -1

Crash course in index theory

Index theorem (Gül, Pearce, and Stacchetti, 1993; Ritzberger, 1994)

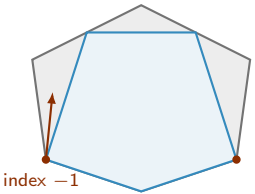
For a regular game, $\sum_{\nu \in \mathrm{NE}} \mathrm{index}(\nu) = +1$

one can also assign an index to equilibrium components when regularity fails, but we won't need it

Corollaries

- $\#\{\text{mixed NE}\} \geq \#\{\text{pure NE}\} - 1$ (Gül, Pearce, and Stacchetti, 1993)
- ≥ 2 NE $\Rightarrow$ some equilibrium has index -1
- if ν has index -1, then ν is mixed and there is another equilibrium $\nu' \neq \nu$

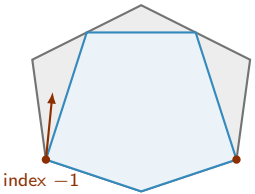
Constructing a CE outside the hull



≥ 2 equilibria, at most two mixers everywhere:

- pick an equilibrium ν of index -1

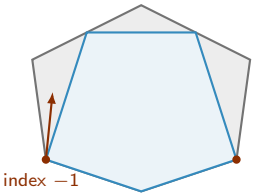
Constructing a CE outside the hull



≥ 2 equilibria, at most two mixers everywhere:

- pick an equilibrium ν of index -1
- remove pure players and actions outside the support, then reverse payoffs $u \mapsto -u$

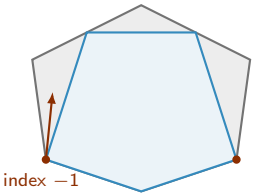
Constructing a CE outside the hull



≥ 2 equilibria, at most two mixers everywhere:

- pick an equilibrium ν of index -1
- remove pure players and actions outside the support, then reverse payoffs $u \mapsto -u$
- index stays -1

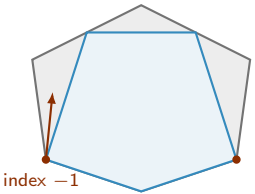
Constructing a CE outside the hull



≥ 2 equilibria, at most two mixers everywhere:

- pick an equilibrium ν of index -1
- remove pure players and actions outside the support, then reverse payoffs $u \mapsto -u$
- index stays -1
- index $-1 \Rightarrow$ the reversed game has another equilibrium ν', with $\mathrm{supp}(\nu') \subseteq \mathrm{supp}(\nu)$

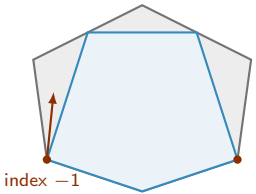
Constructing a CE outside the hull



≥ 2 equilibria, at most two mixers everywhere:

- pick an equilibrium ν of index -1
- remove pure players and actions outside the support, then reverse payoffs $u \mapsto -u$
- index stays -1
- index $-1 \Rightarrow$ the reversed game has another equilibrium ν', with $\mathrm{supp}(\nu') \subseteq \mathrm{supp}(\nu)$
- containment is strict: a regular two-player NE is unique on its support

Constructing a CE outside the hull



≥ 2 equilibria, at most two mixers everywhere:

- pick an equilibrium ν of index -1
- remove pure players and actions outside the support, then reverse payoffs $u \mapsto -u$
- index stays -1
- index $-1 \Rightarrow$ the reversed game has another equilibrium ν', with $\mathrm{supp}(\nu') \subseteq \mathrm{supp}(\nu)$
- containment is strict: a regular two-player NE is unique on its support

For sufficiently small $\varepsilon > 0$, define

$$\mu = \frac{\nu - \varepsilon\,\nu'}{1 - \varepsilon}$$

Choose ε so $\mu \geq 0$ and deviations outside the support remain unprofitable
Then $\mu \in \mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE})$

▸ Why outside

Example: subtracting an anti-Nash equilibrium

A coordination game

	A	B
A	$(1,1)$	$(0,0)$
B	$(0,0)$	$(1,1)$

Consider the NE ν with
$\nu_1 = \nu_2 = (1/2, 1/2)$

$\nu = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has index -1

Reversing payoffs makes (A, B) a pure NE

Example: subtracting an anti-Nash equilibrium

A coordination game

	A	B
A	$(1,1)$	$(0,0)$
B	$(0,0)$	$(1,1)$

Consider the NE ν with
$\nu_1 = \nu_2 = (1/2, 1/2)$

$\nu = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has index -1

Reversing payoffs makes (A, B) a pure NE

Subtract and renormalize

$$\nu' = \delta_{(A,B)}, \qquad \varepsilon = \tfrac{1}{4}$$

$$\mu = \frac{\nu - \varepsilon\nu'}{1-\varepsilon} = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

- $\mu \in \mathrm{CE}$
- every Nash mixture has equal off-diagonal masses
- $\mu(A, B) = 0 < \mu(B, A)$

$$\Rightarrow \quad \mu \in \mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE})$$

A unique CE

Singleton CE

skip to conclusions

Beyond a few structured classes, little is known about games with a unique CE

- concave potential games (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025), strictly socially concave games (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015; Einy, Haimanko, and Lagziel, 2022), zero-sum (Forges, 1990)

Singleton CE

skip to conclusions

Beyond a few structured classes, little is known about games with a unique CE

- concave potential games (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025), strictly socially concave games (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015; Einy, Haimanko, and Lagziel, 2022), zero-sum (Forges, 1990)

We can say more about uniqueness of the coarse analogue, CCE

$$\mathrm{CCE} = \{\nu\} \quad \Longrightarrow \quad \mathrm{CE} = \{\nu\} \qquad (\mathrm{NE} \subseteq \mathrm{CE} \subseteq \mathrm{CCE})$$

Singleton CE

skip to conclusions

Beyond a few structured classes, little is known about games with a unique CE

- concave potential games (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025), strictly socially concave games (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015; Einy, Haimanko, and Lagziel, 2022), zero-sum (Forges, 1990)

We can say more about uniqueness of the coarse analogue, CCE

$$\mathrm{CCE} = \{\nu\} \quad \Longrightarrow \quad \mathrm{CE} = \{\nu\} \qquad (\mathrm{NE} \subseteq \mathrm{CE} \subseteq \mathrm{CCE})$$

Plan

1. Characterize games with a unique CCE (Sandomirskiy and Wincelberg, 2026)
2. Outline what changes for a unique CE (work in progress)

Coarse CE

CCE (Hannan, 1957; Moulin and Vial, 1978)

$\mu \in \Delta(A)$ is a CCE if for every player i and every ex-ante deviation $a_i' \in A_i$

$$\mathbb{E}_\mu\left[u_i(a)\right] \;\geq\; \mathbb{E}_\mu\left[u_i(a_i', a_{-i})\right]$$

- the deviation is before seeing the recommendation $\Rightarrow$ CCE $\supseteq$ CE
- models ex-ante opt-in: collusive agreements, delegation to an intermediary such as a platform or a system of AI agents

Coarse CE

CCE (Hannan, 1957; Moulin and Vial, 1978)

$\mu \in \Delta(A)$ is a CCE if for every player i and every ex-ante deviation $a_i' \in A_i$

$$\mathbb{E}_\mu\left[u_i(a)\right] \;\geq\; \mathbb{E}_\mu\left[u_i(a_i', a_{-i})\right]$$

- the deviation is before seeing the recommendation $\Rightarrow$ CCE $\supseteq$ CE
- models ex-ante opt-in: collusive agreements, delegation to an intermediary such as a platform or a system of AI agents

Dominance $\Rightarrow$ uniqueness: a strictly dominant profile is the unique NE, CE, and CCE

Coarse CE

CCE (Hannan, 1957; Moulin and Vial, 1978)

$\mu \in \Delta(A)$ is a CCE if for every player i and every ex-ante deviation $a_i' \in A_i$

$$\mathbb{E}_\mu\left[u_i(a)\right] \;\geq\; \mathbb{E}_\mu\left[u_i(a_i', a_{-i})\right]$$

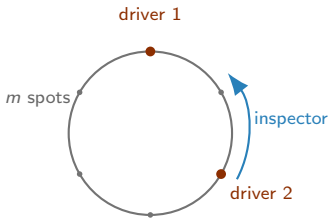
- the deviation is before seeing the recommendation $\Rightarrow$ CCE $\supseteq$ CE
- models ex-ante opt-in: collusive agreements, delegation to an intermediary such as a platform or a system of AI agents

Dominance $\Rightarrow$ uniqueness: a strictly dominant profile is the unique NE, CE, and CCE

What else?

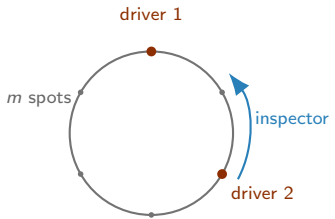
Intuition: a parking game

- 2 drivers: park illegally at one of m spots on a loop, or park legally (cost c)
- exactly one illegal $\Rightarrow$ caught for sure (ticket $t > c$)
- both illegal $\Rightarrow$ inspector starts at a uniformly random point, goes counter-clockwise; only the first found is ticketed



Intuition: a parking game

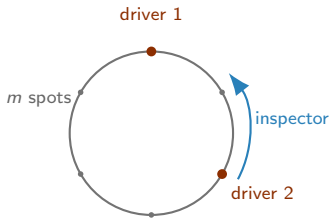
- 2 drivers: park illegally at one of m spots on a loop, or park legally (cost c)
- exactly one illegal $\Rightarrow$ caught for sure (ticket $t > c$)
- both illegal $\Rightarrow$ inspector starts at a uniformly random point, goes counter-clockwise; only the first found is ticketed



- a violator wants to park right after the other $\Rightarrow$ caught with probability only $1/m$
- so legal is strictly dominant only when $\frac{t}{m} > c$

Intuition: a parking game

- 2 drivers: park illegally at one of m spots on a loop, or park legally (cost c)
- exactly one illegal $\Rightarrow$ caught for sure (ticket $t > c$)
- both illegal $\Rightarrow$ inspector starts at a uniformly random point, goes counter-clockwise; only the first found is ticketed



- a violator wants to park right after the other $\Rightarrow$ caught with probability only $1/m$
- so legal is strictly dominant only when $\frac{t}{m} > c$
- yet legal is the unique CCE already when $\frac{t}{2} > c$
- **why:** under any joint violation at least one driver is caught with probability $\geq \frac{1}{2} \Rightarrow$ wants to deviate to legal ex-ante

Enforcement games and unique CCE

A game v is an **enforcement game** if there is $a^* \in A$ such that

$$v_i(a_i^*, a_{-i}) = 0, \quad \forall i, a_{-i}, \qquad \sum_i v_i(a) < 0 \quad \forall a \neq a^*$$

- playing the action a_i^* guarantees 0, whatever others do
- every other profile is negative-sum

Enforcement games and unique CCE

A game v is an **enforcement game** if there is $a^* \in A$ such that

$$v_i(a_i^*, a_{-i}) = 0, \quad \forall i, a_{-i}, \qquad \sum_i v_i(a) < 0 \quad \forall a \neq a^*$$

- playing the action a_i^* guarantees 0, whatever others do
- every other profile is negative-sum

Theorem (Sandomirskiy and Wincelberg, 2026)

A game u has a unique CCE, which is pure $\iff$ strategically equivalent to an enforcement game

Enforcement games and unique CCE

A game v is an **enforcement game** if there is $a^* \in A$ such that

$$v_i(a_i^*, a_{-i}) = 0, \quad \forall i, a_{-i}, \qquad \sum_i v_i(a) < 0 \quad \forall a \neq a^*$$

- playing the action a_i^* guarantees 0, whatever others do
- every other profile is negative-sum

Theorem (Sandomirskiy and Wincelberg, 2026)

A game u has a unique CCE, which is pure $\iff$ strategically equivalent to an enforcement game

- strategic equivalence: $u_i(a_i, a_{-i}) = \gamma_i\big(v_i(a_i, a_{-i}) + \beta_i(a_{-i})\big), \qquad \gamma_i > 0$

Enforcement games and unique CCE

A game v is an **enforcement game** if there is $a^* \in A$ such that

$$v_i(a_i^*, a_{-i}) = 0, \quad \forall i, a_{-i}, \qquad \sum_i v_i(a) < 0 \quad \forall a \neq a^*$$

- playing the action a_i^* guarantees 0, whatever others do
- every other profile is negative-sum

Theorem (Sandomirskiy and Wincelberg, 2026)

A game u has a unique CCE, which is pure $\iff$ strategically equivalent to an enforcement game

- strategic equivalence: $u_i(a_i, a_{-i}) = \gamma_i\big(v_i(a_i, a_{-i}) + \beta_i(a_{-i})\big), \qquad \gamma_i > 0$
- **Proof idea**
  - enforcement $\Rightarrow$ unique CCE:
    $\mu(A \setminus \{a^*\}) > 0 \Rightarrow$ expected welfare $< 0 \Rightarrow \mathbb{E}_\mu[u_i] < 0$ for some $i \Rightarrow i$ would deviate to a_i^*
  - converse: LP duality

Beyond pure uniqueness

Unique CCE **can be mixed** Matching pennies: the unique NE is the unique CCE

	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

Beyond pure uniqueness

Unique CCE **can be mixed** Matching pennies: the unique NE is the unique CCE

	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

Sandomirskiy and Wincelberg (2026)

If $\mathrm{CCE}(G)$ is a singleton, then either it is a pure NE, or exactly two players mix over two actions each and the induced 2×2 game is of matching-pennies type

Beyond pure uniqueness

Unique CCE **can be mixed** Matching pennies: the unique NE is the unique CCE

	H	T
H	$(1,-1)$	$(-1,1)$
T	$(-1,1)$	$(1,-1)$

Sandomirskiy and Wincelberg (2026)

If $\mathrm{CCE}(G)$ is a singleton, then either it is a pure NE, or exactly two players mix over two actions each and the induced 2×2 game is of matching-pennies type

- in symmetric games, uniqueness $\Rightarrow$ purity
- idea: a unique CCE must be an extreme point; richer mixing destroys extremality

From CCE to CE uniqueness

Rudov, Sandomirskiy, and Yariv (2025)

A unique CE has at most two randomizing players

From CCE to CE uniqueness

Rudov, Sandomirskiy, and Yariv (2025)

A unique CE has at most two randomizing players

- extreme-point argument: a unique CE is an extreme point of the CE polytope; ≥ 3 mixers destroy extremality
- a unique CE is quasi-strict (Viossat, 2008) $\Rightarrow$ no regularity / quasi-strictness assumption is needed

From CCE to CE uniqueness

Rudov, Sandomirskiy, and Yariv (2025)

A unique CE has at most two randomizing players

- extreme-point argument: a unique CE is an extreme point of the CE polytope; ≥ 3 mixers destroy extremality
- a unique CE is quasi-strict (Viossat, 2008) $\Rightarrow$ no regularity / quasi-strictness assumption is needed

≤ 2 mixers $\Rightarrow$ the unique CE is pure, or two players mix

we focus on the pure case

A pure unique CE (a preliminary result)

- CCE: each player enforces the target payoff by a single-shot deviation to a_i^*
- CE: enforcement uses a sequence of deviations toward a_i^*

A pure unique CE (a preliminary result)

- CCE: each player enforces the target payoff by a single-shot deviation to a_i^*
- CE: enforcement uses a sequence of deviations toward a_i^*
- consider an enforcement game v with a^* and chains X_i on A_i absorbed at a_i^*
- define cumulative payoff along the chain of deviations:

$$w_i(a_i, a_{-i}) = \mathbb{E}\Big[\sum_{t \geq 0} v_i(X_i^t, a_{-i}) \;\Big|\; X_i^0 = a_i \Big]$$

A pure unique CE (a preliminary result)

- CCE: each player enforces the target payoff by a single-shot deviation to a_i^*
- CE: enforcement uses a sequence of deviations toward a_i^*
- consider an enforcement game v with a^* and chains X_i on A_i absorbed at a_i^*
- define cumulative payoff along the chain of deviations:

$$w_i(a_i, a_{-i}) = \mathbb{E}\Big[\sum_{t \geq 0} v_i(X_i^t, a_{-i}) \;\Big|\; X_i^0 = a_i \Big]$$

Proposition

A game u has a unique CE, which is pure $\iff$ strategically equivalent to w (for some v, X)

A pure unique CE (a preliminary result)

- CCE: each player enforces the target payoff by a single-shot deviation to a_i^*
- CE: enforcement uses a sequence of deviations toward a_i^*
- consider an enforcement game v with a^* and chains X_i on A_i absorbed at a_i^*
- define cumulative payoff along the chain of deviations:

$$w_i(a_i, a_{-i}) = \mathbb{E}\Big[\textstyle\sum_{t\geq 0} v_i(X_i^t, a_{-i}) \;\Big|\; X_i^0 = a_i \Big]$$

Proposition

A game u has a unique CE, which is pure $\iff$ strategically equivalent to w (for some v, X)

- the gap CE vs CCE: one-shot vs multistage chains
- future: Economic interpretation? Interesting examples?

Conclusion

Conclusion

Economic takeaway

- mediation, communication, and learning bring new outcomes relative to decentralized play whenever it leaves anything to coordinate on

Nerdy takeaway

- a nonlinear object, the NE, its complex counterpart, and index govern the local structure of a linear object, the CE polytope
- more surprising links likely to be found

Conclusion

Economic takeaway

- mediation, communication, and learning bring new outcomes relative to decentralized play whenever it leaves anything to coordinate on

Nerdy takeaway

- a nonlinear object, the NE, its complex counterpart, and index govern the local structure of a linear object, the CE polytope
- more surprising links likely to be found

Thank you!

Why three mixers prevent extremality: an example

- n players, each with two actions

Why three mixers prevent extremality: an example

- n players, each with two actions
- Any CE μ satisfies the incentive constraints

$$\sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i', a_{-i})$$

Why three mixers prevent extremality: an example

- n players, each with two actions
- Any CE μ satisfies the incentive constraints

$$\sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i', a_{-i})$$

- $2n$ constraints

Why three mixers prevent extremality: an example

- n players, each with two actions
- Any CE μ satisfies the incentive constraints

$$\sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i', a_{-i})$$

- $2n$ constraints
- Winkler (1988): if k linear constraints are imposed on the set of all distributions $\Delta(A)$, extreme distributions have support $\leq k + 1$

Why three mixers prevent extremality: an example

- n players, each with two actions
- Any CE μ satisfies the incentive constraints

$$\sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \mu(a_i, a_{-i}) u_i(a_i', a_{-i})$$

- $2n$ constraints
- Winkler (1988): if k linear constraints are imposed on the set of all distributions $\Delta(A)$, extreme distributions have support $\leq k + 1$
- $\Rightarrow$ an extreme CE has support size at most $2n + 1$

Why three mixers prevent extremality: an example

- Suppose ν is a NE with k mixers

Why three mixers prevent extremality: an example

- Suppose ν is a NE with k mixers
- ν has 2^k action profiles in its support

Why three mixers prevent extremality: an example

- Suppose ν is a NE with k mixers
- ν has 2^k action profiles in its support
- $\Rightarrow$ For ν to be extreme,

$$2^k \leq 2n + 1$$

Why three mixers prevent extremality: an example

- Suppose ν is a NE with k mixers
- ν has 2^k action profiles in its support
- $\Rightarrow$ For ν to be extreme,

$$2^k \leq 2n + 1$$

- Removing pure players sharpens the bound to

$$2^k \leq 2k + 1$$

Why three mixers prevent extremality: an example

- Suppose ν is a NE with k mixers
- ν has 2^k action profiles in its support
- $\Rightarrow$ For ν to be extreme,

$$2^k \leq 2n + 1$$

- Removing pure players sharpens the bound to

$$2^k \leq 2k + 1$$

Conclusion: a NE with $k \geq 3$ mixers cannot be extreme

Why three mixers prevent extremality: an example

- Suppose ν is a NE with k mixers
- ν has 2^k action profiles in its support
- $\Rightarrow$ For ν to be extreme,

$$2^k \leq 2n + 1$$

- Removing pure players sharpens the bound to

$$2^k \leq 2k + 1$$

Conclusion: a NE with $k \geq 3$ mixers cannot be extreme

- The same argument applies to equilibria where players mix over the same number of pure strategies

Why three mixers prevent extremality: an example

- Suppose ν is a NE with k mixers
- ν has 2^k action profiles in its support
- $\Rightarrow$ For ν to be extreme,

$$2^k \leq 2n + 1$$

- Removing pure players sharpens the bound to

$$2^k \leq 2k + 1$$

Conclusion: a NE with $k \geq 3$ mixers cannot be extreme

- The same argument applies to equilibria where players mix over the same number of pure strategies
- The difficulty: very asymmetric supports

Back

General case, for totally-mixed regular NE

General support bound: player i faces $|A_i|(|A_i|-1)$ incentive constraints
By Winkler (1988), every extreme CE satisfies

$$\#\,\mathrm{supp}(\mu) \le 1 + \textstyle\sum_i |A_i|(|A_i|-1)$$

General case, for totally-mixed regular NE

General support bound: player i faces $|A_i|(|A_i|-1)$ incentive constraints
By Winkler (1988), every extreme CE satisfies

$$\#\,\mathrm{supp}(\mu) \le 1 + \textstyle\sum_i |A_i|(|A_i|-1)$$

This bound alone does not rule out product distributions: with action-set sizes $(4,2,2)$,

$$|A_1||A_2||A_3| = 4\cdot 2\cdot 2 = 16 \;\le\; 1+12+2+2 = 17$$

General case, for totally-mixed regular NE

General support bound: player i faces $|A_i|(|A_i|-1)$ incentive constraints
By Winkler (1988), every extreme CE satisfies

$$\#\,\mathrm{supp}(\mu) \le 1 + \textstyle\sum_i |A_i|(|A_i|-1)$$

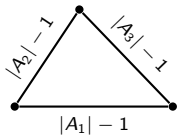
This bound alone does not rule out product distributions: with action-set sizes $(4,2,2)$,

$$|A_1||A_2||A_3| = 4\cdot 2\cdot 2 = 16 \;\le\; 1+12+2+2 = 17$$

Such examples are ruled out by the polygon inequality
For a regular NE ν

$$|A_i| - 1 \le \sum_{j\neq i} (|A_j|-1) \qquad \forall i$$

- the total number of degrees of freedom of $j \neq i$ is at least the number of indifference conditions of i



Triangle inequality when $n = 3$

$(4,2,2)$ violates $3 \le 1+1$

General case, for totally-mixed regular NE

General support bound: player i faces $|A_i|(|A_i|-1)$ incentive constraints
By Winkler (1988), every extreme CE satisfies

$$\#\,\mathrm{supp}(\mu) \le 1 + \textstyle\sum_i |A_i|(|A_i|-1)$$

This bound alone does not rule out product distributions: with action-set sizes $(4,2,2)$,

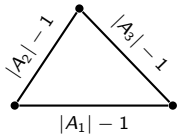
$$|A_1||A_2||A_3| = 4\cdot 2\cdot 2 = 16 \;\le\; 1+12+2+2 = 17$$

Such examples are ruled out by the polygon inequality
For a regular NE ν

$$|A_i| - 1 \le \sum_{j\neq i} (|A_j|-1) \qquad \forall i$$

- the total number of degrees of freedom of $j \neq i$ is at least the number of indifference conditions of i

Together: these bounds rule out extremality with ≥ 3 mixers



Triangle inequality when $n = 3$

$(4,2,2)$ violates $3 \le 1+1$

Back

Appendix: why another algebraic root exists

McKelvey and McLennan (1997):
With $d_i = |\operatorname{supp}(\nu_i)| - 1 \geq 1$ (pure players omitted), the generic number of complex Nash equilibria is

$$B(d_1, \ldots, d_k) = [t_1^{d_1} \cdots t_k^{d_k}] \prod_i \left(\sum_{j \neq i} t_j \right)^{d_i}$$

- a nonsingular support Jacobian requires $d_i \leq \sum_{j \neq i} d_j$ for every i
- $\Rightarrow B(d) \geq (k-1)!$
- with 3+ mixers, these inequalities imply $B(d) \geq 2$
- binary mixers: $B(1,1,1) = 2$ and $B(1,1,1,1) = 9$

◂ Construction

Appendix: why the anti-Nash perturbation escapes the Nash hull

Consider $\mu = \dfrac{\nu - \varepsilon\,\nu'}{1-\varepsilon}$, where ν' the anti-NE with $\mathrm{supp}(\nu')$ strictly inside $\mathrm{supp}(\nu)$

- $\Rightarrow \nu'$ omits some action $c_i \in \mathrm{supp}(\nu_i)$ for a player $i \in \{1, 2\}$
- hence the player-i marginal increases at c_i:

$$\mu_i(c_i) = \frac{\nu_i(c_i)}{1-\varepsilon} > \nu_i(c_i)$$

Appendix: why the anti-Nash perturbation escapes the Nash hull

Consider $\mu = \dfrac{\nu - \varepsilon\,\nu'}{1-\varepsilon}$, where ν' the anti-NE with $\mathrm{supp}(\nu')$ strictly inside $\mathrm{supp}(\nu)$

- $\Rightarrow$ ν' omits some action $c_i \in \mathrm{supp}(\nu_i)$ for a player $i \in \{1,2\}$
- hence the player-i marginal increases at c_i:

$$\mu_i(c_i) = \frac{\nu_i(c_i)}{1-\varepsilon} > \nu_i(c_i)$$

- suppose for contradiction $\mu = \sum_\ell \lambda_\ell\,\nu^\ell$ with $\nu^\ell \in \mathrm{NE}$, $\lambda_\ell > 0 \Rightarrow \mathrm{supp}(\nu^\ell) \subseteq \mathrm{supp}(\nu)$
- at μ, conditional on c_i, player i is indifferent across $\mathrm{supp}(\nu_i)$
- $\Rightarrow$ the same holds under each $\nu^\ell \Rightarrow \mathrm{supp}(\nu_i^\ell) = \mathrm{supp}(\nu_i) \Rightarrow \mathrm{supp}(\nu^\ell) = \mathrm{supp}(\nu)$ for all ℓ
- in two-player games, no regular NE can share support with another NE, so every $\nu^\ell = \nu$
- then $\mu_i(c_i) = \nu_i(c_i)$, contradiction

no such decomposition exists $\Rightarrow$ $\mu \in \mathrm{CE} \setminus \mathrm{conv}(\mathrm{NE})$

back

References I

Itai Ashlagi, Dov Monderer, and Moshe Tennenholtz. On the value of correlation. Journal of Artificial Intelligence Research, 33:575–613, 2008

Robert J Aumann. Subjectivity and correlation in randomized strategies. Journal of mathematical Economics, 1(1):67–96, 1974

Robert J Aumann. Correlated equilibrium as an expression of bayesian rationality. Econometrica: Journal of the Econometric Society, pages 1–18, 1987

Imre Bárány. Fair distribution protocols or how the players replace fortune. Mathematics of Operations Research, 17(2):327–340, 1992

Elchanan Ben-Porath. Correlation without mediation: Expanding the set of equilibrium outcomes by "cheap" pre-play procedures. Journal of Economic Theory, 80(1):108–122, 1998

Antoni Calvó-Armengol. The set of correlated equilibria of 2x2 games. mimeo, 2006

Sabrina Gomez Canovas, Pierre Hansen, and Brigitte Jaumard. Nash equilibria from the correlated equilibria viewpoint. International Game Theory Review, 1(01):33–44, 1999

Zhigang Cao, Zhibin Tan, and Jinchuan Zhou. Correlated equilibrium of games with concave potential functions. Operations Research Letters, 60:107241, 2025

M Cripps. Extreme correlated and Nash equilibria in two-person games, 1995

Ezra Einy, Ori Haimanko, and David Lagziel. Strong robustness to incomplete information and the uniqueness of a correlated equilibrium. Economic Theory, 73(1):91–119, 2022

Fe S Evangelista and TES Raghavan. A note on correlated equilibrium. International Journal of Game Theory, 25:35–41, 1996

Eyal Even-Dar, Yishay Mansour, and Uri Nadav. On the convergence of regret minimization dynamics in concave games. In Proceedings of the forty-first annual ACM symposium on Theory of computing, pages 523–532, 2009

Michal Feldman, Brendan Lucier, and Noam Nisan. Correlated and coarse equilibria of single-item auctions. In Web and Internet Economics: 12th International Conference, WINE 2016, Montreal, Canada, December 11-14, 2016, Proceedings 12, pages 131–144. Springer, 2016

References II

Françoise Forges. Correlated equilibrium in two-person zero-sum games. Econometrica (1986-1998), 58(2):515, 1990

Dean P Foster and Rakesh V Vohra. Calibrated learning and correlated equilibrium. Games and Economic Behavior, 21(1-2):40–55, 1997

Drew Fudenberg and David K Levine. Conditional universal consistency. Games and Economic Behavior, 29(1-2):104–130, 1999

Dino Gerardi. Unmediated communication in games with complete and incomplete information. Journal of Economic Theory, 114(1): 104–131, 2004

Faruk Gül, David Pearce, and Ennio Stacchetti. A bound on the proportion of pure strategy equilibria in generic games. Mathematics of Operations Research, 18(3):548–552, 1993

James Hannan. Approximation to bayes risk in repeated play. Contributions to the Theory of Games, 3(2):97–139, 1957

John C Harsanyi. Oddness of the number of equilibrium points: a new proof. International Journal of Game Theory, 2:235–250, 1973

Sergiu Hart and Andreu Mas-Colell. A simple adaptive procedure leading to correlated equilibrium. Econometrica, 68(5):1127–1150, 2000

Sergiu Hart and Andreu Mas-Colell. Markets, correlation, and regret-matching. Games and Economic Behavior, 93:42–58, 2015

Atsushi Kajii and Stephen Morris. The robustness of equilibria to incomplete information. Econometrica, 65(6):1283–1309, 1997. ISSN 00129682, 14680262. URL http://www.jstor.org/stable/2171737

Navin Kartik, Francesco Squintani, and Katrin Tinn. Information revelation in constant-sum games: Elections and beyond. arXiv preprint arXiv:2406.17084, 2024

Deniz Kattwinkel, Axel Niemeyer, Justus Preusser, and Alexander Winter. Mechanisms without transfers for fully biased agents. In Proceedings of the 23rd ACM Conference on Economics and Computation, pages 332–333, 2022

Kiumars Kaveh and A. G. Khovanskii. A short survey on Newton polytopes, tropical geometry and ring of conditions of algebraic torus, 2018. URL https://arxiv.org/abs/1803.07001

References III

Frédéric Koessler, Marco Scarsini, and Tristan Tomala. Correlated equilibria in large anonymous bayesian games. Mathematics of Operations Research, 50(3):2157–2174, 2025

Ehud Lehrer and Sylvain Sorin. One-shot public mediated talk. Games and Economic Behavior, 20(2):131–148, 1997

Richard D McKelvey and Andrew McLennan. The maximal number of regular totally mixed Nash equilibria. Journal of Economic Theory, 72(2):411–425, 1997

Hervé Moulin and J P Vial. Strategically zero-sum games: the class of games whose completely mixed equilibria cannot be improved upon. International Journal of Game Theory, 7:201–221, 1978

Abraham Neyman. Correlated equilibrium and potential games. International Journal of Game Theory, 26:223–227, 1997

Christos H Papadimitriou and Tim Roughgarden. Computing correlated equilibria in multi-player games. Journal of the ACM (JACM), 55 (3):1–29, 2008

Gregory Pavlov. Correlated equilibria and communication equilibria in all-pay auctions. Review of Economic Design, pages 1–33, 2023

Klaus Ritzberger. The theory of normal form games from the differentiable viewpoint. International Journal of Game Theory, 23(3): 207–236, 1994. doi: 10.1007/BF01247316

Kirill Rudov, Fedor Sandomirskiy, and Leeat Yariv. Extreme equilibria: The benefits of correlation. In Proceedings of the 26th ACM Conference on Economics and Computation, pages 897–897, 2025

Fedor Sandomirskiy and Ben Wincelberg. Delegation in strategic environments and equilibrium uniqueness. arXiv preprint arXiv:2602.21470, 2026

Takashi Ui. Correlated equilibrium and concave games. International Journal of Game Theory, 37:1–13, 2008

Yannick Viossat. The geometry of Nash equilibria and correlated equilibria and a generalization of zero-sum games. Technical report, SSE/EFI Working Paper Series in Economics and Finance, 2006

Yannick Viossat. Is having a unique equilibrium robust? Journal of Mathematical Economics, 44(11):1152–1160, 2008

Gerhard Winkler. Extreme points of moment sets. Mathematics of Operations Research, 13(4):581–587, 1988